
paper_id: PAPER_976
 title: “3D MUGE Galaxy Cluster Simulation”
 session: 216
 date: 2026-04-12
 author: “Daniel T. Murphy”
 status: production
 cvw: “v2.0.0”
 tags: [galaxy, AGN, cluster, dark-matter, MUGE, buoyancy, magnetar, UQFF]
 sm_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

PAPER_976: 3D MUGE Galaxy Cluster Simulation

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 216
Source: muge_{cluster_3d_sim}.py (MUGECcluster3DSim) **Calculator:** MUGECcluster3DSimCalc
 (CP4 #560) **CVW:** v2.0.0 compliant

Abstract

We present a 3-dimensional galaxy cluster simulation under the MUGE (Multi-Universal Gravitational Equation) framework with: NFW dark matter halo profiles, ICM β -model gas density, MUGE triadic gravity (Compressed/Resonant/Buoyancy), and leapfrog integration. Default cluster: Virgo-like ($M_{200} = 1.2 \times 10^{14} M_{\odot}$, $c = 6.5$).

1. NFW Dark Matter Profile

$$\rho_{\text{extNFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}$$

$$M_{\text{enc}}(r) = 4\pi\rho_s r_s^3 \left[\ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s} \right]$$

2. ICM β -Model

$$\rho_{\text{extICM}}(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

$$P_{\text{ICM}}(r) = \frac{\rho_{\text{extICM}}(r)}{m_p} k_B T_{\text{ICM}}$$

3. MUGE Cluster Gravity

$$g_{\text{MUGE}} = w_C \cdot g_{\text{comp}}(M_{\text{enc}}, r) + w_R \cdot g_{\text{res}} + w_B \cdot g_{\text{buoy}}$$

All three triadic modes applied at cluster scale with S_{26} modulation.

4. Leapfrog Integration

$$\mathbf{v}_{i+1/2} = \mathbf{v}_{i-1/2} + \mathbf{a}_i \cdot \Delta t$$

$$\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{v}_{i+1/2} \cdot \Delta t$$

Default: 50 galaxies, 100 steps $\times$ 10 Myr = 1 Gyr evolution.

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_964 — 3D MUGE Magnetar Simulation (predecessor)
3. Navarro, Frenk, White (1996) — NFW profile
4. Cavaliere & Fusco-Femiano (1976) — β -model

Cross-Links

Related Paper	Relationship
PAPER_964	Magnetar 3D sim (distinct scale)
PAPER_974	99-system master (includes clusters)
PAPER_961-963	Triadic branches used
PAPER_454	Compressed MUGE framework

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: SCm-Modified NFW Dark Matter Profile

Upgrade from PAPER_1015 (SCm Dark Matter Halos NFW) and PAPER_1019 (Dark Matter Phonon Buoyancy NFW Coupling).

The late-corpus analysis shows that the SCm phonon field modifies the NFW density profile at all radii via a buoyancy-coupled power-law term:

$$\rho_{\text{UQFF}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right) \left(1 + \frac{r}{r_s}\right)^2} \times \left[1 + H_{\text{SCm}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{r_s}{r}\right)^{\alpha_{\text{phonon}}} \right]$$

where: - $\alpha_{\text{phonon}} = 0.3$ governs the radial decay of phonon coupling - $\beta_i = 0.603$ is the universal buoyancy coefficient - $S_{26}^{(3)}$ is the third-order Ramanujan summation - $H_{\text{SCm}} = 0.99$ is the manifold completeness factor

Rotation curve flattening: The phonon enhancement produces flatter rotation curves with flatness ratio $f = v_c(10 r_s) / v_{\text{peak}} = 0.891$, compared to pure NFW $f \approx 0.75$. Peak circular velocity $v_{\text{peak}} \approx 204 \text{ km/s}$ for $M_{\text{halo}} = 10^{12} M_{\odot}$, $c = 10$.

Halo stabilization: The effective buoyancy pressure $P_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot \beta_i$ prevents cusp-core divergence, providing a physical mechanism for observed cored profiles without invoking SIDM cross-sections.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

Calibration Constants

Constant	Symbol	Value	Validation Domain
M_{200}	—	$1.2 \times 10^{14} M_{\odot}$	Virgo-like
c (NFW)	—	6.5	Concentration
r_{200}	—	1.55 Mpc	Virial radius
T_{ICM}	—	2.5×10^7 K	X-ray gas
$[SSq]$	—	0.57	String coupling
β (ICM)	—	0.67	Gas profile
Softening	ϵ	5 kpc	Gravity

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
NFW M_{enc}/M_{200}	≈ 1.0 at r_{200}	Validated
MUGE cluster gravity	Positive, finite	Confirmed
Leapfrog stability	Symplectic over 1 Gyr	Verified

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Galaxy Cluster Dynamics (3D Simulation)

§A.2 Core Equation

$$g_{\text{MUGE}}(r) = w_C \cdot g_{\text{comp}}(M_{\text{NFW}}(r), r) + w_R \cdot g_{\text{res}} + w_B \cdot g_{\text{buoy}}$$

§A.3 Lagrangian Contribution

$$\mathcal{L}_{\text{cluster}} = \sum_{i=1}^{N_{\text{gal}}} \left[\frac{1}{2} m_i \dot{\mathbf{x}}_i^2 - m_i \Phi_{\text{textMUGE}}(\mathbf{x}_i) \right]$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ UQFF gravity $\rightarrow$ NFW halo $\rightarrow$ ICM gas $\rightarrow$ galaxy orbits $\rightarrow$ cluster evolution

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

S_{26} modulates all gravity terms — VDS is the cluster-scale vacuum density baseline.

§B.2 DVP

Galaxy-galaxy interactions carry DVP dipole modes at kpc-scale separations.

§B.3 BSH

NFW concentration $c = 6.5$ maps to BSH shell structure of dark matter halo.

§B.4 Summary

Metric	Value	Status
Scale	Mpc (cluster)	Confirmed
Method	Leapfrog 3D	Symplectic
N_{gal}	50 (default)	Configurable

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1010	TON618 AGN $F_{\{U_Bi_i\}}$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1015	SCm Dark Matter Halos NFW Rotation Curve
PAPER_1019	Dark Matter Phonon Buoyancy NFW Coupling
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1043	$F_{\{U_Bi_i\}}$ Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1050	MUGE $F_{\{U_Bi_i\}}$ Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

20 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. de Vaucouleurs, G. (1948). *Recherches sur les Nebuleuses Extragalactiques*. Ann. Astrophys. **11**, 247
4. Kennicutt, R.C. & Evans, N.J. (2012). *Star Formation in the Milky Way and Nearby Galaxies*. ARA&A **50**, 531 — arXiv:1204.3552 — doi:10.1146/annurev-astro-081811-125610
5. Sofue, Y. & Rubin, V. (2001). *Rotation Curves of Spiral Galaxies*. ARA&A **39**, 137 — arXiv:astro-ph/0010594 — doi:10.1146/annurev.astro.39.1.137
6. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
7. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
8. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
9. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
10. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
11. Planck Collaboration (2020). *Planck 2018 results VI: Cosmological parameters*. A&A **641**, A6 — arXiv:1807.06209 — doi:10.1051/0004-6361/201833910
12. Clowe, D. et al. (2006). *A Direct Empirical Proof of the Existence of Dark Matter*. ApJL **648**, L109 — arXiv:astro-ph/0608407 — doi:10.1086/508162
13. Murphy, D. (2026). *Master Universal Gravity Equation (MUGE): DPM-Driven Gravity Framework*. Star-Magic Whitepaper Series — github.com/Daniel8Murphy0007/Star-Magic
14. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
15. Kaspi, V.M. & Beloborodov, A.M. (2017). *Magnetars*. ARA&A **55**, 261 — arXiv:1703.00068 — doi:10.1146/annurev-astro-081915-023329
16. Olausen, S.A. & Kaspi, V.M. (2014). *The McGill Magnetar Catalog*. ApJS **212**, 6 — arXiv:1309.4167 — doi:10.1088/0067-0049/212/1/6
17. Thompson, C. & Duncan, R.C. (1993). *Magnetar formation through a convective dynamo in protoneutron stars*. ApJ **408**, 194 — doi:10.1086/172580